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General Instructions: Write Program in C++ to solve the following problems with and **without one or more user defined functions where appropriate**, as stated in the individual question.

Problem No.	SL No.	Description	Page No.
P-1	1. a)	Solve suppressed cubic equations or transcendental functions by using bisection & Newton-Raphson method.	
P-2	1. b)	Solve cubic general cubic equation by using C++ programming.	
P-3	2. a)	Calculate the value of $\sin x$, $\cos x$ and $\log(1 + x)$ using its series for some given x using WHILE or DO WHILE condition.	
P-4	2. b)	Program to find the value of $\sin x$ using its series for some given x using WHILE or DO WHILE condition.	
P-5	2. c)	Program to find the value of $\cos x$ using its series for some given x using WHILE or DO WHILE condition.	
P-6	3. a)	Program to check whether any given number is prime or not. Write codes for both with and without user defined function.	
P-7	3. b)	Program to find the factorial of any given number. Write codes for both with and without user defined function.	
P-8	3. c)	Program to sort an array of numbers in ascending and descending order. Write codes for both with and without user defined function.	
P-9	3. d)	Program to find the largest and smallest number that can be formed using digits of a given number. Write codes for both with and without user defined function.	
P-10	4. a)	Program to take input of a user name and consumed current units and print an electric bill which billing criteria is as follows i) First 50 units has minimum charge 100tk ii) Next 200 units cost 2.50 TK/unit iii) Next 250 units cost 3.50 TK/unit iv) Units above 500 are charged at a rate 5.00TK/unit Make sure you implement the idea using user-defined function.	
P-11	4. b)	Program to take input of a user name and yearly income and print income tax where the taxation criteria is as follows i) First 250000 is tax free ii) 5% tax for next 250000 iii) 10% tax for next 500000 iv) 20% tax for next 4000000 v) 40% tax for income above 5000000 Make sure you implement the idea using user-defined function.	
P-12	4. c)	Program to take input of a user name and minutes spent talking on the telephone and print a telephone bill which billing criteria is as follows i) First 150 minutes is free of cost ii) Next 250 calls (151-400 minute) are charged at the rate of 1 TK/minute iii) And all calls after 400 minutes, are charged at the rate of 2/minute Make sure you implement the idea using user-defined function.	
P-13	4. d)	Program to take input of the name an of third year student of Applied Mathematics, RU and marks obtained in all courses including LAB and	

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		viva, and print the overall GPA in year three following the standard grading criteria of Applied Mathematics.	
P-14	5. a)	Program to find the dominant eigenvalue and corresponding eigenvector of a given non-diagonal square matrix.	
P-15	5. b)	Program to find the inverse of a given square matrix using row-elementary operations, and verify your answer.	
P-16	5. c)	Program to find the determinant of square matrix using row-elementary operations.	
P-17	6. a)	Program to solve a system of n linear equations by using Gauss's Elimination method.	
P-18	6. b)	Program to solve a system of n linear equations by using Cramer's rule.	
P-19	6. c)	Program to find the smallest and largest distance between any two points taken from a randomly given set of n points.	